

Business Requirements Document (BRD)

1. Document Overview

1.1 Purpose

This Business Requirements Document (BRD) defines the **business and operational requirements** for the VidyaTrack system.

The objective is to ensure accurate attendance tracking, academic performance recording, and reliable parent communication for a single school pilot.

1.2 Scope

This BRD covers:

- Attendance management
- Marks and performance recording
- Parent communication
- Management oversight

This document does **not** cover:

- System architecture
 - UI/UX design
 - Technical implementation details
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2. Business Context

2.1 School Profile

- Pilot Scope: Single school
 - School Type: Primary and Secondary
 - Medium of Instruction: English
 - Internet Reliability: Intermittent
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2.2 Business Objectives

- Reduce manual effort in attendance and marks tracking
 - Ensure timely communication with parents
 - Provide management with accurate student data on demand
 - Preserve academic records independent of teacher turnover
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3. Stakeholders

Stakeholder	Interest
School Management	Visibility and accountability
Teachers	Simple daily academic recording
Admin Staff	Operational data management
Parents	Timely student updates

4. User Roles (Business Definition)

4.1 Teacher

- Subject-based teachers
- Each teacher independently records attendance and marks for assigned classes
- Responsible for accuracy and timeliness of academic data

4.2 Admin

- Uses a separate portal
- Manages operational and master data
- Does not enter academic data

4.3 Management

- Uses a separate portal
- View-only access to academic data
- Can send custom messages to parents

4.4 Parent

- No system login
 - Receives notifications only (WhatsApp and SMS)
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5. Functional Requirements

5.1 Attendance Management

BR-ATT-01

The system shall allow teachers to mark attendance **once per day per class**.

BR-ATT-02

The system shall allow teachers to edit attendance **only on the same day of submission**.

BR-ATT-03

Attendance shall become read-only after the same-day edit window closes.

BR-ATT-04

The system shall send:

- Absence notifications
 - Daily attendance summaries to parents
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5.2 Marks & Academic Performance

BR-MRK-01

The system shall support the following exam types:

- Unit tests
- Mid-term exams
- Final exams

BR-MRK-02

The system shall allow both:

- Numeric marks
- Grade-based evaluation

BR-MRK-03

Teachers must enter **mandatory remarks** when submitting marks.

BR-MRK-04

Marks submission without remarks shall not be allowed.

BR-MRK-05

Parents shall receive **consolidated exam results** per exam cycle.

5.3 Parent Communication

BR-COM-01

The system shall send notifications via:

- WhatsApp
- SMS

BR-COM-02

The system shall generate automated messages for:

- Attendance
- Exam results

BR-COM-03

Management shall be able to send **custom messages** to parents.

5.4 Management Access

BR-MGT-01

Management shall have **view-only access** to:

- Attendance records
- Marks and remarks
- Student profiles

BR-MGT-02

Management shall not be allowed to edit attendance or marks.

6. Business Workflows (Textual)

6.1 Attendance Workflow

1. Teacher selects class and subject
 2. Marks attendance for the day
 3. Submits attendance
 4. System confirms submission
 5. Parent notifications are sent
 6. Attendance may be edited only on the same day
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6.2 Marks Entry Workflow

1. Teacher selects exam type
 2. Enters marks for all students
 3. Enters mandatory remarks
 4. Submits marks
 5. System validates completeness
 6. Consolidated results are sent to parents
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7. Business Rules

- Attendance can be marked only once per day per class
 - Attendance edits allowed only on the same day
 - Marks cannot be submitted without remarks
 - Parents receive consolidated exam results only
 - Management access is strictly view-only
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8. Edge Cases & Exceptions

- Intermittent internet may delay submission
 - Duplicate attendance submissions must be prevented
 - Notification failures must attempt alternate channel (WhatsApp → SMS)
 - Partial data entry must be blocked
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9. Non-Functional Requirements (Business Level)

- The system must be usable by teachers with low to medium technical proficiency
 - Data must remain accurate and auditable
 - System must tolerate intermittent connectivity
 - Records must persist beyond teacher tenure
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10. Assumptions & Constraints

Assumptions

- Teachers will complete entries on the same day
- Parent contact details are available and accurate

Constraints

- Single-school pilot
 - No parent portal access
 - Internet reliability is not guaranteed
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11. Approval Criteria

The system will be considered acceptable for pilot use when:

- Attendance and marks can be recorded without errors
- Parents consistently receive notifications
- Management can retrieve student information on demand
- Teachers can operate the system with minimal training